

Guillaume Moinard | Postdoc in Computer Science

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Academic Background

Postdoc.....

Complexity Science Hub

Digital Currency Ecosystems Team

Vienna

2026–2027

Topic: Decentralized Governance in the Ğ1 ecosystem

Supervisor: Bernhard Haslhofer

Abstract: I study governance within a decentralized economy: the Ğ1 currency. Among other things, I explore how distributed this ecosystem actually is and whether the web of trust, a decentralized identification system, efficiently prevents fraud without preventing newcomers from joining this economic network.

PhD.....

LIP6

Sorbonne Université - CNRS

Paris

2022–2026

Topic: Walkers scattered in Street Networks: Robust Decentralized Gatherings

Supervisor: Matthieu Latapy

Abstract: How to meet if we are dispersed in a city? Coordinating, by exchanging our positions thanks to modern means of communication, seems easy. But how to achieve the same result if communications are impossible and no meeting point is identifiable? This is the case, for example, during a repressed social movement or a blackout. The objective of my thesis is to find the key ingredients for a movement tactic to ensure that walkers will meet in the same place, regardless of their starting positions. I therefore model them as random walks, on a street network, biased by various observables [5]. Having identified sufficient rules to guarantee the formation of such groups, I subsequently improved this model, and obtained more effective tactics, by incorporating a more realistic vision in the walkers [4]. Finally, in the case of countermeasures or crowd movements, I show that these walks offer guarantees regarding the reformation of groups after a dispersion [3].

Education

Sorbonne Université

Paris

Master 2 Physics of Complex Systems, Grade: Very Good

2021–2022

Courses in dynamical systems, machine learning, econophysics, stochastic processes, game theory, statistical field theory and graph theory.

Sorbonne Université

Clermont-Ferrand

Gap Year

2020–2021

Internship at INRAE, tutoring in a high school, and associative project of solidarity distributions as vice-president at the Maison Des Solidarités Étudiantes.

University of Durham

Durham, England

Master 1 in Physics, Grade: Good

2019–2020

Erasmus exchange with Sorbonne Université. Participation in the *DurHack* hackathon.

Sorbonne Université

Paris

Bachelor in Physics, Grade: Good

2016–2019

Thesis Prizes

Bertalanffy Young Doctoral Award 🌐

Bordeaux

Finalist, French Regional Conference on Complex Systems

2025

Finalist of a competition for which I presented my thesis work through a 5-minute video.

My Thesis in 180 Seconds 🌐

Paris

Finalist, Sorbonne Université

2025

Finalist of an eloquence competition for which I presented my thesis topic in 3 minutes.

Summer Schools

Lake Como School of Advanced Studies

Como, Italy

Complex Networks: Theory, Methods, and Applications

2023

Courses on graph models, epidemiology phenomena in networks, statistical comparison tools for graphs, and introduction to hypergraphs.

Internships

M2 Internship with Matthieu Latapy and Nicolas Maudet

Paris

LIP6, 5 months

2022

Study of new forms of protests. Implementation of multi-agent models governed by stochastic processes on city graphs obtained by extracting data from OpenStreetMap.

M1 Internship with Guillaume Deffuant

Clermont-Ferrand

INRAE, 5 months

2021

Participation in the ToRealSim project. Extension, in C++, of the multi-agent opinion propagation model *Léviathan* to social networks.

L3 Internship with Antoine Moreau

Clermont-Ferrand

Institut Pascal, 1 month

2018–2019

Optimization of the computation time of the *MoosH* software simulating multi-layers for wave optics in Matlab. Study of *Differential Evolution*, an optimization algorithm inspired by natural selection.

Publications

Research Topics: Network Science, Urban Networks, Random Walks, Multi-Agent Systems, Collective Intelligence, Computational Social Sciences

International Conference Proceedings with Peer Review.....

- [2] Guillaume Moinard and Matthieu Latapy. “BeWater: Effective Protesters Navigate Watersheds in Street Networks”. In: *The 25th International Conference on Autonomous Agents and Multi-Agent Systems*. 2025. DOI: 10.65109/EIDE2300
- [4] Guillaume Moinard and Matthieu Latapy. “Improving flocking behaviors in street networks with vision”. In: *Complex Networks XVI: Proceedings of the 16th Conference on Complex Networks*. Springer. 2025. DOI: 10.1007/978-3-031-93619-7_10
- [5] Guillaume Moinard and Matthieu Latapy. “Fast Flocking of Protesters on Street Networks”. In: *ASONAM: International Conference on Advances in Social Networks Analysis and Mining*. Springer. 2024, pp. 194–203. DOI: 10.1007/978-3-031-78538-2_17
- [6] Etienne Russeil, Fabricio Olivetti de Francca, Konstantin Malanchev, Bogdan Burlacu, Emille Ishida, Marion Leroux, Clement Michelin, Guillaume Moinard, and Emmanuel Gangler. “Multi-view Symbolic Regression”. In: *GECCO: Proceedings of the Genetic and Evolutionary Computation Conference*. 2024, pp. 961–970. DOI: 10.1145/3638529.3654087

Journal Articles.....

- [1] Etienne Russeil, Fabricio Olivetti de França, Konstantin Malanchev, Guillaume Moinard, and Maxime Cherrey. “Exploring Multi-view Symbolic Regression methods in physical sciences”. In: *Phil. Trans. R. Soc. A* (2026). DOI: 10.1098/rsta.2024.0592
- [3] Guillaume Moinard and Matthieu Latapy. “Fast and Robust Flocking of Protesters on Street Networks”. In: *submitted* (2025). URL: <https://hal.science/hal-04594953>

Software Contributions.....

libSG

Library, Pierre Baumann and Guillaume Moinard 2024

Library written in C to process temporal graphs. The library offers a data structure to represent the abstract type of Stream Graphs developed by Matthieu Latapy. With this method, it is possible to define primitives and efficiently calculate the metrics introduced in the article in question.

POG

Program, Ivan Mulot-Radojcic and Guillaume Moinard 2023

Simulator of walks on street networks in C associated with an article [5] on the gathering of walkers on street graphs. Project presented in the form of a companion site to explain how to use the code.

Presentations

Conferences without Proceedings with Peer Review.....

International Network for Social Network Analysis	Paris
<i>Sunbelt "Social Networks, Mechanisms, and Algorithms"</i>	2025
BeWater: Effective Protesters Navigate Watersheds in Street Networks	
Institut des Systèmes Complexes	Paris
<i>Bridging Theoretical and Empirical approaches for the study of Social Dynamics</i>	2024
Flocking of Protesters on Street Networks	
Institut National des sciences appliquées	Lyon
<i>Archipel</i>	2024
Le Dernier Kilomètre	
University of Pennsylvania	Philadelphia, USA
<i>International Conference on Computational Social Science</i>	2023
Gathering and Flocking of Protesters on Street Networks	
Université Le Havre Normandie	Le Havre
<i>French Regional Conference on Complex Systems</i>	2023
Impact of pedestrian flocking tactics on urban networks	

Workshop.....

Sciences Po	Toulouse
<i>Colloque inter-AtÉcoPol</i>	2025
Organizer and animator of the workshop <i>Recherches pour les luttes</i> . A workshop in the format of a world café to develop research dedicated to social and ecological struggles.	
KTH Royal Institute of Technology	Stockholm, Sweden
<i>ICT4S</i>	2024
Creation of a video to present the concept of the "inverse Moore's law".	

Seminars.....

ISIR	Paris
<i>ACIDE Team Seminar</i>	2025
Marches aléatoires dans les réseaux de rues : vers des rassemblements totalement décentralisés	
LIRIS	Lyon
<i>DM2L Team Seminar</i>	2025
BeWater : Vers des Rassemblements Décentralisés Efficaces Dans les Réseaux de Rues	
LPENS	Lyon
<i>Sisyphé Team Seminar</i>	2025
BeWater : Vers des Rassemblements Décentralisés Efficaces Dans les Réseaux de Rues	
LIFAT	Tours
<i>Team seminar Reconnaissance des Formes & Analyse d'Images</i>	2025
BeWater : Vers des Rassemblements Décentralisés Efficaces Dans les Réseaux de Rues	
LIASD	Saint-Denis
<i>Seminar ouverture en informatique et de formation of the PASTIS team</i>	2025
BeWater : Vers des Rassemblements Décentralisés Efficaces Dans les Réseaux de Rues	
Centre Georges Pompidou	Paris
<i>Day of the École Doctorale Informatique, Télécommunications et Électronique</i>	2025
Presentation of my PhD work with the my thesis in 180 seconds format.	

LIP6	Paris
<i>PhD student day for the A.I. & Data Science laboratory research axis</i>	2024
Flocking and Be Water: Robust Tactics for Protests on Street Networks	
Doctoral Collective of Sorbonne Université	Paris
<i>Sorbonne Actuelle</i>	2024
Doctorant.e face à son sujet : comment politiser sa recherche.	
LIP6	Paris
<i>Séminaire des doctorant.e.s de la SIF</i>	2024
Flocking Tactics for Protests on Street Networks	
Inter-team Colloquium	Arc-et-Senans
<i>Project CNRS - Mission pour les Initiatives Transverses et Interdisciplinaires</i>	2023
Flocking Tactics for Protests on Street Networks	
Lake Como School of Advanced Studies	Como
<i>Complex Networks: Theory, Methods, and Applications, Italy</i>	2023
Flocking Tactics for Protests on Street Networks	
Symposium on Academia and Activism	Fribourg
<i>Université de Fribourg, Switzerland</i>	2022
Presentation of the results of my M2 internship, as well as the research perspectives of my thesis.	

Posters.....

Sorbonne Université	Paris
<i>Yearly day of LIP6</i>	2024
Modélisation et Analyse de Traces de Mobilités	
Sorbonne Université	Paris
<i>Yearly day of LIP6</i>	2023
Robustesse et Fragilité des Réseaux d'Infrastructures	
Sorbonne Université	Paris
<i>Day of the École Doctorale Informatique, Télécommunications et Électronique</i>	2023
Robustesse et Fragilité des Réseaux d'Infrastructures	

Teaching

Supervision of Internships.....

Sorbonne Université	Paris
<i>Co-supervision, 2-month internship</i>	2024
Supervision of Pierre Baumann in a L3 internship, with Matthieu Latapy, to implement a library of Stream Graph functions in C.	
Sorbonne Université	Paris
<i>Co-supervision, 3-month internship</i>	2023
Supervision of Ivan Mulot-Radojcic in a L2 internship, with Matthieu Latapy, to optimize random walk programs in C.	

Courses.....

UFR d'Ingénierie de Sorbonne Université	Paris
<i>Doctoral Teaching Mission, 193.5 hours</i>	2022-2025

- During my teaching, I notably had the responsibility of teaching part of the M2 *Graph And Networks* lecture course, for which I chose the pedagogical content and prepared the slides. I completed my course by creating a set of Anki 🧠 flashcards to help students in their revisions.
- I also participated in the correction of exam papers and that of codes submitted in practical work. I wrote the subjects of several continuous assessments for the tutorials.

Course	Level	Format	Volume	Content
<i>Introduction to Programming</i>	L1	TP	39.5h	Learning Python. Presentation of functions, loops, and data structures.
<i>Supervised Research Project</i>	L1	Project	28h	Programming project of dynamical systems and discovery of the bibliographic method.
<i>Introduction to Programming</i>	L1	TD	35h	Learning Python. Presentation of functions, loops, and data structures.
<i>Algorithms</i>	L2	TD	43.5h	Discrete mathematics, proofs of termination and validity of classical algorithms.
<i>Research Project</i>	L3	Project	12h	Supervision of a group around the study of an issue on temporal graphs.
<i>Advanced C</i>	L2	TP	12.25h	Deepening of concepts such as pointers, structures, strings, the libC.
<i>Graphs and Networks (course in English)</i>	M2	CM	5h	Course on propagation phenomena in real networks (epidemics, rumors, etc).
<i>Data Structures</i>	L2	TP	19.25h	Programming of data structures in C. Implementation of a CPU simulator.

Institution Saint-Alyre

Tutoring, 2 years ≈ 200h

Clermont-Ferrand

2017 et 2020

Weekly tutoring sessions in Physics-Chemistry, provided to high school students, set up within the institution, in 3-hour slots. Additional private lessons in the evenings for a few students.

Scientific Outreach

Conference cycle on les mystères de l'Autre

Speaker

Université Populaire de Bagnolet

2025

Connecting with Others: Understanding Networks, with Matthieu Latapy.

Stand Up For Science

Animator

Paris

2025

Introduction to computer science for passersby in a Parisian place, with a mediation workshop on algorithms: the *psychorigid pancake maker*, where the goal is to successfully sort pancakes, during the scientific outreach event on April 3.

La fête de la science

Animator

Sorbonne Université

2024

Introduction to computer science for primary school classes with a mediation workshop on sorting algorithms: the *psychorigid pancake maker* 🧠.

Paris en Science

Writer

Sorbonne Université

2023

Writer of the quantum physics section for a scientific outreach booklet distributed in high schools and on the campuses of Sorbonne Université by the student association Connectome In Science.

Scientific Outreach Competition	Sorbonne Université
<i>Winner</i>	2022
Winner of the scientific eloquence competition of the association Connectome In Science.	
OpenRadiation	Festival d'Astronomie de Fleurance
<i>Animator</i>	2022
Demonstrations of radioactivity measurements made with the sensor of the OpenRadiation association and promotion of open science to the participants, general public, of the festival in partnership with OpenRadiation.	
Journal Alma Mater	Sorbonne Université
<i>Writer</i>	2018–2019
Writing articles in a monthly inter-university student newspaper, both in print and online. Distribution of newspapers and writing articles on science and climate. Editor-in-chief for one issue.	

Collective Responsibilities

Peer Review

Applied Network Science, IC2S2, KDD, CompleNet, WSDM, TMA, ASONAM, FRCCS, CIKM	
<i>Sub - Reviewer</i>	2022-2025
Peer review of articles, delegated by members of the editorial committee of several conferences.	

Event Organization

Seminar of ComplexNetworks	LIP6
<i>Team Seminars</i>	2024-2025
Invitation of speakers: <i>Pablo Rauzy, Erwan Taillanter</i> . Organization of events, reservation of the restaurant and transportation, animation of the seminar.	
Sorbonne Actuelle	Sorbonne Université
<i>Interdisciplinary Seminar</i>	2024
Organization of the interdisciplinary seminar on the theme "Doctorant·e face à son sujet" with the Collectif Doctoral de Sorbonne Université.	
Colloquium at the Saline Royale	Arc-et-Senans
<i>Perturbation des Infrastructures en Réseaux par des Mouvements Sociaux</i>	2023
Organization of seminars over 3 days between several teams as part of the CNRS project <i>MITI-PRIMS</i> on the study of social movements. Invitation of speakers: César Ducruet, Julien Randon-Furling, Claire Lagesse. Preparation of a schedule, reservation of the venue, and animation of the colloquium.	

Institutional Representation

Complexity Science Hub	Vienna
<i>Postdoc Representative</i>	2026-2027
Member of the CSH Faculty council, and coordinator of Postdoc events.	
Doctoral College	Sorbonne Université
<i>Elected PhD Student</i>	2025-2027
Elected on the list of the collectif doctoral to link the different doctoral schools, which make up the doctoral college of Sorbonne Université, and the PhD students. Advocacy for the introduction of VSS training for all thesis supervisors.	
Laboratory Council	LIP6
<i>Elected PhD Student</i>	2025
Member of the laboratory council. Intermediary between the PhD students and the laboratory management.	

Council of the Faculty of Sciences and Engineering

Elected PhD Student

Sorbonne Université

2023-2025

Elected on the list of Solidaires Étudiant·e·s. Link between the faculty of sciences and the students of this faculty. Organization of a survey on the issue of integral continuing education. Active participation in the assembly of student representatives.

Council of PhD Students

Team Representative

LIP6

2022-2025

Discussions with the laboratory management, welcoming new students. Interview with the HCÉRES committee to represent the PhD students. Arrangement of a PhD student room.

Council of the UFR of Physics

Elected Student

Sorbonne Université

2021-2023

Link between the Physics students and the UFR. Vote on the allocations of the UFR budget. Participation in the rewriting of the internal regulations.

Scientific Associative Commitments.....

Sciences Citoyennes

Administration council member

2025-2026

Member of the association's board and of the working group on a proposal for a constitutional law to establish citizens' conventions. Promoting democratic decision-making in science, by proposing that part of the research budget be allocated to these citizens' conventions. Recruitment of a doctoral student for a thesis co-financed with the ANRT.

CurieOsity

Vice-President

Sorbonne Université

2021-2022

Association, under the 1901 law, of physics students. Organizer of the master's forum, the M2 graduation ceremony, conferences with Jean-Philippe Bouchaud, with Francis Rocard and professor coffee breaks. Animator of tutoring sessions and astrophotography evenings. Organizer of the trip to the 2022 edition of the Fleurance Festival. In charge of managing partnerships.

Débattre en Sorbonne

Member

Inter-University

2018-2023

Association recognized as being of public interest, inter-university of oratory art. Responsible for the links between the association and the administration of Sorbonne Université. Organization of a conference with Étienne Klein. Writing and declamation of a eulogy of Marie Curie at the Panthéon.

Dédale

President

Université Clermont Auvergne

2017-2018

During my first year of bachelor's degree, completed in Clermont-Ferrand, creation of Dédale, collective of physics students at the Université Clermont Auvergne. Writing of course notes in collaboration with the teachers. Sharing these on the university's moodle site.